

Unit 2 Guided Notes — Exponents & Scientific Notation

Same base? Move the exponents. Big or tiny number? Pack it into a $\times 10^n$.

Standards: NC.8.EE.1 · NC.8.EE.3 · NC.8.EE.4

This packet has three parts: **Part 1 — Guided Notes**, **Part 2 — Practice Problems**, and **Part 3 — Answer Key**. Fill the blanks on your printed copy first, then click any blank, step, or check button to reveal the answer.

Topics Covered: 2.1 Exponent Rules — Product, Quotient, Power, Zero, Negative (NC.8.EE.1) · 2.2 Scientific Notation — Format & Conversions (NC.8.EE.3) · 2.2 Multiplying & Dividing Scientific Notation (NC.8.EE.4) · 2.2 Word Problems with Scientific Notation (NC.8.EE.4)

1 Exponent Rules

✦ **KEY RULE:** Exponent rules only apply when the BASES are the SAME!

Fill the examples and key phrases on your printed copy, then click to check.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Product Rule	$x^a \times x^b = x^{a+b}$			
Quotient Rule	$x^a \div x^b = x^{a-b}$			
Power Rule	$(x^a)^b = x^{a \times b}$			
Zero Exponent	$x^0 = 1$			
Negative Exponent	$x^{-n} = 1/x^n$			

⚠ PARENTHESES MATTER!

$$-3^2 = \text{ } \text{ vs. } (-3)^2 = \text{ }$$

Without (), ONLY the base gets the exponent — not the negative sign!

PRODUCT RULE: COUNT THE FACTORS

$$\underbrace{\boxed{x} \boxed{x} \boxed{x}}_{x^3 \text{ (3 factors)}} \times \underbrace{\boxed{x} \boxed{x}}_{x^2 \text{ (2)}} = \underbrace{\boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x}}_{x^5 \text{ (5 factors)}}$$

$$x^3 \times x^2 = x^{3+2} = x^5 \rightarrow \text{ADD the exponents}$$

Modeling the Rules — See WHY

Each rule below comes from just writing out the repeated multiplication. Click a card to model it with factors — **first base**, **second base**, **result**.

Product Rule · $x^3 \times x^2 = x^5$

See why →

$$\boxed{x} \boxed{x} \boxed{x} \times \boxed{x} \boxed{x} = \boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} = \boxed{x^5}$$

Count the factors: $3 + 2 = 5$. Same base → ADD the exponents.

Quotient Rule · $x^5 \div x^2 = x^3$

See why →

$$\boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} \div \boxed{x} \boxed{x} \rightarrow \text{cancel 2 pairs} \rightarrow \boxed{x} \boxed{x} \boxed{x} = \boxed{x^3}$$

Cancel matching factors: $5 - 2 = 3$. Same base → SUBTRACT the exponents.

Power Rule · $(x^2)^3 = x^6$

See why →

$$\boxed{x^2} \boxed{x^2} \boxed{x^2} = \boxed{x} \boxed{x} \cdot \boxed{x} \boxed{x} \cdot \boxed{x} \boxed{x} = \boxed{x^6}$$

Three copies of x^2 : $2 \times 3 = 6$. Power to a power → MULTIPLY.

Zero Exponent · $x^0 = 1$

See why →

$$\boxed{x} \boxed{x} \boxed{x} \div \boxed{x} \boxed{x} \boxed{x} = 1, \text{ but by the quotient rule } = x^{3-3} = x^0. \text{ So } \boxed{x^0 = 1}$$

Anything (except 0) divided by itself is 1 → $x^0 = 1$.

Negative Exponent · $x^{-3} = 1/x^3$

See why →

$$\boxed{x} \boxed{x} \div \boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} = x^{2-5} = x^{-3}; \text{ cancelling leaves } \boxed{1 / x^3}$$

A negative exponent means 'flip it' → $x^{-n} = 1/x^n$.

Quick Check

MC Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?

✓ Same base, multiplication → Product Rule: ADD the exponents. $x^5 \cdot x^3 = x^8$.

Match the Rule to its Key Phrase

MATCH Click a rule, then its key phrase.

Rule

Product Rule

Quotient Rule

Power Rule

Negative Exponent

Key Phrase

Power to power? MULTIPLY

Same base? ADD exponents

Flip to a fraction

Same base? SUBTRACT exponents

0 of 4 matched

Worked Example — Simplify $(4x^3y^2)(3x^5y)$

Step 1 – Multiply coefficients: $4 \times 3 = 12$

Step 2 – Add x exponents: $x^3 \cdot x^5 = x^8$

Step 3 – Add y exponents: $y^2 \cdot y^1 = y^3$

✓ Step 4 – Combine: $12x^8y^3$

ERROR ANALYSIS Spot the Error — Simplify $(3x^2)^3$. A student wrote $3x^6$.

The coefficient 3 must ALSO be cubed: $3^3 = 27$. Correct: $27x^6$. The power rule applies to EVERYTHING inside the parentheses.

2 Scientific Notation

$a \times 10^n$ where $1 \leq a < 10$

4.5×10^7 | coefficient · base (always 10) · exponent

SCIENTIFIC NOTATION: MOVE THE DECIMAL

$$45,000 \rightarrow 4.5 \times 10^4$$

big number → move LEFT 4 places → exponent +4

$$0.0067 \rightarrow 6.7 \times 10^{-3}$$

small number → move RIGHT 3 places → exponent -3

Type	Direction	Exponent	Example
Large number	Move decimal LEFT		$45,000 = 4.5 \times 10^4$
Small number (< 1)	Move decimal RIGHT		$0.0067 = 6.7 \times 10^{-3}$

Multiplying Scientific Notation — 3-Step Process

1. the coefficients (the a-values)
2. the exponents
3. Adjust if coefficient is NOT between and

Example (no adjustment): $(3 \times 10^4) \times (2 \times 10^5)$

Multiply coefficients: $3 \times 2 = 6$

ADD exponents: $4 + 5 = 9$

✓ Answer: 6×10^9

Dividing Scientific Notation — 3-Step Process

1. the coefficients
2. the exponents
3. Adjust if needed

Example (no adjustment): $(8 \times 10^6) \div (4 \times 10^2)$

Divide coefficients: $8 \div 4 = 2$

SUBTRACT exponents: $6 - 2 = 4$

✓ Answer: 2×10^4

⚠ ADJUSTMENT RULES:

- If coefficient ≥ 10 : Move decimal LEFT, ADD 1 to exponent
- If coefficient < 1 : Move decimal RIGHT, SUBTRACT 1 from exponent

Example needing adjustment: $(5 \times 10^3) \times (4 \times 10^2)$

Multiply: $5 \times 4 = 20$

Add exponents: $3 + 2 = 5 \rightarrow \text{result: } 20 \times 10^5$

Adjust ($20 \geq 10$): $20 = 2.0 \times 10^1$, so ADD 1 to exponent

✓ **Final answer:** 2.0×10^6

ERROR ANALYSIS

Spot the Error — Convert 0.0045 to scientific notation. A student wrote 4.5×10^3 .

Small number $\rightarrow$ NEGATIVE exponent. Correct: 4.5×10^{-3} . Numbers less than 1 always have negative exponents in scientific notation.



PART 2 — Practice Problems

Show all work on your printed copy. Circle your final answer, then check on screen.

A Exponent Rules with Monomials

Rules: Multiply $\rightarrow$ ADD exponents | Divide $\rightarrow$ SUBTRACT | Power-to-Power $\rightarrow$ MULTIPLY

1 Simplify $x^3 \cdot x^5$

answer

2 Simplify $(y^4)^3$

answer

3 Simplify $a^8 \div a^3$

answer

4 Simplify $(2x^3)(5x^2)$

answer

5 Simplify $(3m^2)^3$

answer

6 Simplify $12y^7 \div 4y^2$

answer

7 Simplify $(-2a^4)^2$

answer

8 Simplify $(5x^2y)(3xy^3)$

answer

9 Simplify $18n^6 \div 6n^6$

answer

10 Simplify $(4b^3)^2 \div 2b$

answer

11 Simplify $(-3x^2)(-4x^3)$

answer

12 Simplify $(x^5y^2)^3$

answer

B Scientific Notation Conversions

Remember: $a \times 10^n$ where $1 \leq a < 10$. Positive exp = BIG number. Negative exp = SMALL number.

1 45,000,000 → scientific notation

$a \times 10^n$

2 0.00032 → scientific notation

$a \times 10^n$

3 6.02×10^8 → standard form

standard

4 9.1×10^{-5} → standard form

standard

5 780,000 → scientific notation

$a \times 10^n$

6 0.0000067 → scientific notation

$a \times 10^n$

7 3.45×10^4 → standard form

standard

8 1.2×10^{-3} → standard form

standard

9 5,280,000,000 → scientific notation

$a \times 10^n$

10 2.5×10^{-2} → standard form

standard

C Operations

Steps: Multiply/divide the coefficients. Add/subtract the exponents. Adjust if coefficient is not between 1 and 10.

1 $(3 \times 10^4)(2 \times 10^5)$

$a \times 10^n$

2 $(8 \times 10^7) \div (4 \times 10^3)$

$a \times 10^n$

3 $(5 \times 10^3)(7 \times 10^6)$

$a \times 10^n$

4 $(9.6 \times 10^8) \div (3.2 \times 10^5)$

$a \times 10^n$

D Word Problems

Strategy: Identify what operation to use. Write your sci notation setup. Adjust if needed.

1. The Sun is about 9.3×10^7 miles from Earth. Light travels 1.86×10^5 miles per second. How many seconds does it take light to reach Earth?

$$(9.3 \times 10^7) \div (1.86 \times 10^5) = \mathbf{5 \times 10^2 \text{ seconds (500 seconds)}}$$

2. A bacteria cell is about 2×10^{-6} m long. A virus is about 5×10^{-8} m long. How many times longer is the bacteria?

$$(2 \times 10^{-6}) \div (5 \times 10^{-8}) = 0.4 \times 10^2 = \mathbf{40 \text{ times longer}}$$

3. The US national debt is about 3.4×10^{13} dollars. The US population is about 3.4×10^8 people. What is the debt per person?

$$(3.4 \times 10^{13}) \div (3.4 \times 10^8) = \mathbf{1 \times 10^5 = \$100,000 \text{ per person}}$$

4. A grain of sand has mass 6.5×10^{-4} grams. A beach has about 8×10^{18} grains of sand. What is the total mass in grams?

$$(6.5 \times 10^{-4})(8 \times 10^{18}) = 52 \times 10^{14} = \mathbf{5.2 \times 10^{15} \text{ grams}}$$

5. There are about 3.7×10^{13} cells in a human body. Each cell is about 1×10^{-5} m wide. If all cells were lined up, how long in meters?

$$(3.7 \times 10^{13})(1 \times 10^{-5}) = \mathbf{3.7 \times 10^8 \text{ meters}}$$

6. A computer does 4.8×10^9 calculations per second. A task needs 1.44×10^{12} calculations. How many seconds to finish?

$$(1.44 \times 10^{12}) \div (4.8 \times 10^9) = 0.3 \times 10^3 = \mathbf{3 \times 10^2 \text{ seconds (300 seconds)}}$$

★ PART 3 — Answer Key ★

Do not distribute to students until after the lesson.

★ Answers

2.1 Exponent Rules

Product: $x^3 \times x^4 = x^7$; $(2x^3)(3x^2) = 6x^5$; "Same base? ADD." Quotient: $x^6 \div x^2 = x^4$; $(12x^7) \div (3x^2) = 4x^5$; "Same base? SUBTRACT." Power: $(x^2)^3 = x^6$; $(2x^3)^2 = 4x^6$; "Power to power? MULTIPLY." Zero: $5^0 = 1$; $(3x^5)^0 = 1$; "ANY base⁰=1." Negative: $2^{-3} = 1/8$; $4x^{-3} = 4/x^3$; "Flip to fraction." $-3^2 = -9$ vs $(-3)^2 = 9$. Worked: $(4x^3y^2)(3x^5y) = 12x^8y^3$. Spot the error: $(3x^2)^3 = 27x^6$.

2.2 Scientific Notation

Large $\rightarrow$ positive exp; small $\rightarrow$ negative exp. Multiply: MULTIPLY coefficients, ADD exponents, adjust to 1–10. Divide: DIVIDE coefficients, SUBTRACT exponents. $(3 \times 10^4)(2 \times 10^5) = 6 \times 10^9$. $(8 \times 10^6) \div (4 \times 10^2) = 2 \times 10^4$. $(5 \times 10^3)(4 \times 10^2) = 20 \times 10^5 \rightarrow 2.0 \times 10^6$. Spot the error: $0.0045 = 4.5 \times 10^{-3}$.

Part A — Monomials

1) x^8 2) y^{12} 3) a^5 4) $10x^5$ 5) $27m^6$ 6) $3y^5$ 7) $4a^8$ 8) $15x^3y^4$ 9) 3 10) $8b^5$ 11) $12x^5$ 12) $x^{15}y^6$.

Part B — Conversions

1) 4.5×10^7 2) 3.2×10^{-4} 3) 602,000,000 4) 0.000091 5) 7.8×10^5 6) 6.7×10^{-6} 7) 34,500 8) 0.0012 9) 5.28×10^9 10) 0.025.

Part C — Operations

1) 6×10^9 2) 2×10^4 3) 3.5×10^{10} 4) 3×10^3 .

Part D — Word Problems

1) 5×10^2 s (500 s) 2) 40 $\times$ longer 3) $1 \times 10^5 = \$100,000$ 4) 5.2×10^{15} g 5) 3.7×10^8 m 6) 3×10^2 s (300 s).

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